

Kasdin et al. (2025) show that dopamine in the brains of young zebra finches acts as a learning signal, increasing when they sing closer to their adult song and decreasing when they sing further away, effectively guiding their vocal development through trial-and-error. This suggests that complex natural behaviors, like learning to sing, are shaped by dopamine-driven reinforcement learning, similar to how artificial intelligence learns. You can find the paper at this link: <https://www.nature.com/articles/s41586-025-08729-1>.

Note they measure dopamine using fibre photometry, changes in the fluorescence indicate dopamine changes in realtime. Their specific measurement considers changes in fluorescence in 100-ms windows between 200 and 300 ms from the start of singing, averaged across development.

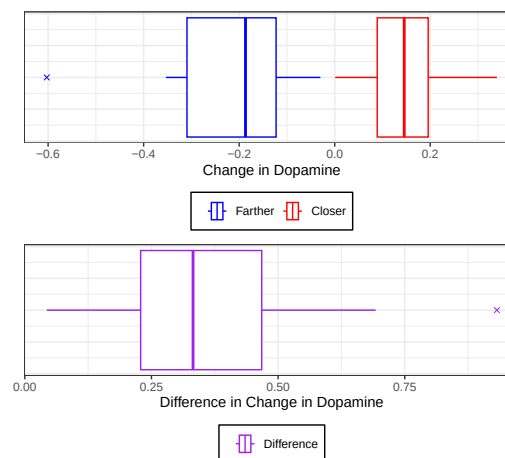
1. Using the **pwr** package for R (Champely, 2020), conduct a power analysis. How many observations would the researchers need to detect a moderate-to-large effect ($d = 0.65$) when using $\alpha = 0.05$ and default power (0.80) for a two-sided one sample t test.

```
(n = pwr.t.test(d = 0.65, sig.level = 0.05, power = 0.80,
               alternative='two.sided', type = 'one.sample')$n)

[1] 20.58039
```

2. Click the link to go to the paper. Find the source data for Figure 2. Download the Excel file. Describe what you needed to do to collect the data for Figure 2(g). Note that you only need the **closer_vals** and **further_vals**. Ensure to **mutate()** the data to get a difference (e.g., **closer_vals** - **further_vals**).
3. Summarize the data.
 - (a) Summarize the further data. Do the data suggest that dopamine in the brains of young zebra finches decreases when they sing further away?
 - (b) Summarize the closer data. Do the data suggest that dopamine in the brains of young zebra finches increases when they sing closer to their adult song?
 - (c) Summarize the paired differences. Do the data suggest that there is a difference between dopamine in the brains of young zebra finches when they sing further away compared to closer to their adult song?

Below are a numerical summary providing a few summary statistics for each data: Closer, Further, and Difference, and a graphical summary displaying box plots for each of the data. All of these summaries suggest that there is a difference between young zebra finches who are closer to their adult song than ones who are not, and that ones who are closer experience a positive change in dopamine while those who are further experience a negative change in dopamine.



	names	mean	sd	variance	kurtosis
1	Closer	0.16	0.09	0.01	-0.86
2	Further	-0.20	0.13	0.02	1.19
3	Difference	0.36	0.21	0.04	0.13

4. Conduct the inferences they do in the paper. Make sure to report the results a little more comprehensively – that is your parenthetical should look something like: ($t = 23.99$, $p < 0.0001$; $g = 1.34$; 95% CI: 4.43, 4.60).

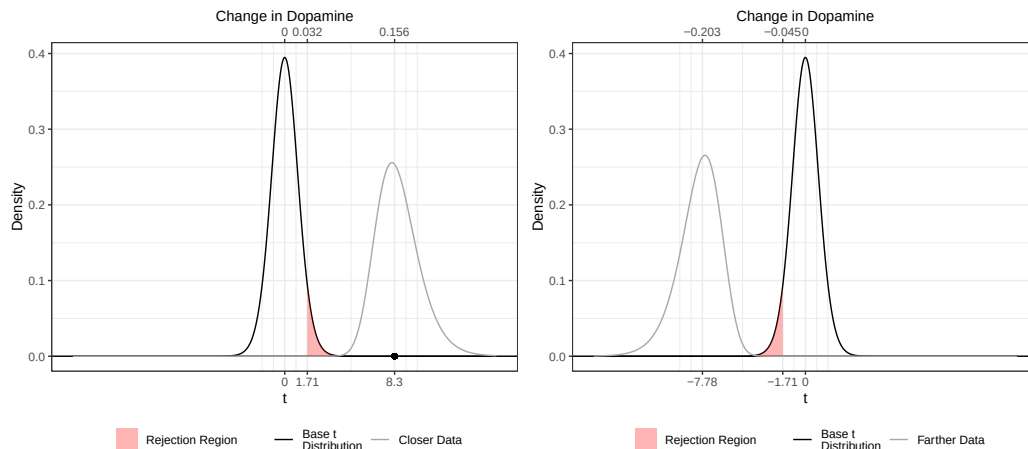
Note: Your numbers may vary slightly as they performed some unclear correction of their p -values. I'm waiting to hear back from them via email!

- (a) “The close responses differed significantly from 0 ($p = 1.63 \times 10^{-8}$).”
- (b) “The far responses differed significantly from 0 ($p = 5.17 \times 10^{-8}$).”
- (c) “The difference between populations was significant ($p = 1.04 \times 10^{-8}$).”

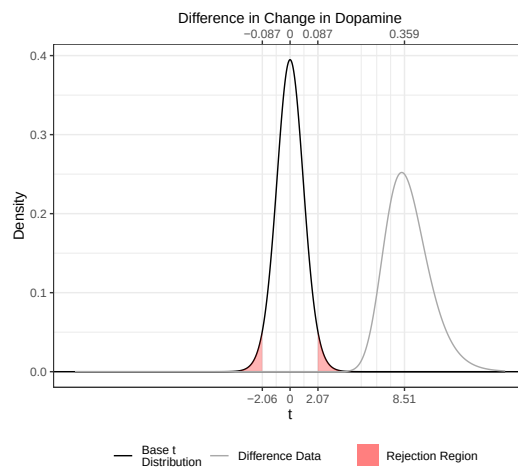
	names	t	p	interval	g
1	Closer	8.30	> 0.00001	[0.155, 0.157]	1.61
2	Further	-7.78	> 0.00001	[-0.204, -0.201]	-1.51
3	Difference	8.51	> 0.00001	[0.356, 0.362]	1.65

5. Reverse engineer the hypothesis test plot from Lecture 20 to create accurate hypothesis testing plots for each part of the previous question.

- (a) Question 4, part(a and b).



(b) Question 4, part(c).



References

Champely, S. (2020). *pwr: Basic Functions for Power Analysis*. R package version 1.3-0.

Kasdin, J., Duffy, A., Nadler, N., Raha, A., Fairhall, A. L., Stachenfeld, K. L., and Gadagkar, V. (2025). Natural behaviour is learned through dopamine-mediated reinforcement. *Nature*, pages 1–8.